# Samuel Phan

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## Education

**University of California, Irvine** Expected March 2028
B.S. Electrical Engineering

## Experience

**CMOS Image Sensor Characterization Intern**, Omnivision – Santa Clara, CA June 2026 – Present

- Identified 20+ implementation differences across Noise, Stream, BLE, and HBand automatic test scripts by comparing HQ and Belgium test flows, algorithms, parameters, and reporting differences
- Validated script differences for product design teams by developing Python A/B tests, smoke tests, and unit tests
- Standardized RMS data aggregation and DN-to-electrons conversion methods by delivering 4 comparison reports

## Projects

**Arctos 6-Axis Robotic Arm** Jan 2026 – Present

- Led project management for a team of 7 members using Notion to track documentation, parts inventory, and project progress
- Delivered 168 printed components within a strict 4kg material budget and ahead of deadline by optimizing strength-to-filament ratios with 4 walls, 15% adaptive cubic infill and staggering print jobs

**Elderly Fall Detection System** Feb 2026

- Created a real-time fall detection system at IrvineHacks to alert caregivers, by integrating OpenCV with an Arduino Uno Q and a webcam
- Detected falls with 83% accuracy and under 12 ms latency by training an image classification model in Edge Impulse using 328 custom scenario images
- Eliminated cloud processing requirements and ensured user privacy by deploying the machine learning model as edge AI on the Arduino Uno Q

**Nike Account Generator** Feb 2021 – Apr 2021

- Developed a Chromium browser automation tool in Go to mass-generate Nike accounts, increasing raffle entry success compared to manual entry methods
- Reduced total account generation time by 10x using goroutines in parallel for concurrent account creation
- Ensured 100% account uniqueness by integrating proxy rotation and SMS API verification

## Extracurriculars

**NASA Community College Aerospace Scholars (NCAS) Missions 1 & 2** Jan 2025 – Apr 2025

- Led research on mechanical systems and mobility solutions for a simulated lunar base, contributing to a team design reviewed by NASA staff
- Advised 13-member team during meetings on budget decisions for mechanical components

## Skills

**Programming:** Python, C++, Verilog, MATLAB, Go
**Design Tools:** LTspice, Vivado, Fusion360 CAD
**Embedded Systems:** Arduino, Raspberry Pi, ESP8266
**Miscellaneous:** Oscilloscope, Function Generator, Multimeter, Linux, ROS, Git